ToyADMOS2: Another dataset of miniature-machine operating sounds for anomalous sound detection under domain shift conditions

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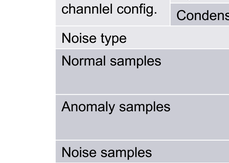
What is ToyADMOS2 ?

Monitoring Task 2

- “ToyADMOS2” is a new large-scale dataset for anomaly detection in machine operating sound
- Designed for evaluating anomaly detection systems under Domain-shift conditions

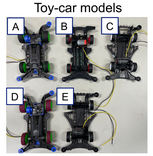
The dataset is freely available at https://github.com/nttcslab/ToyADMOS2-dataset



A subset of this dataset is used in the DCASE2021 Challenge Task2: **Unsupervised Anomalous Sound Detection for Machine Condition Monitoring under Domain Shifted Conditions**

What is ToyADMOS2 ?

Toy-car models



Toy-train models

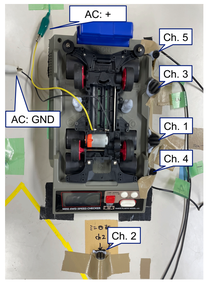
- Collected operating sound of miniature-machines (toys) under normal and anomaly conditions by deliberately damaging them
- with a controlled depth of damages (3 levels) in the anomaly samples

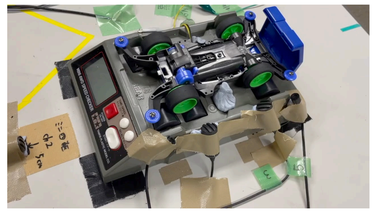
Design choice for domain-shift conditions

Variation settings of sub-datasets

		Toy car	Toy train
Model variations		Five (A, B, C, D, E)	Five (A, B, C, D, E)
Speed levels		Five (1, 2, 3, 4, 5)	Five (1, 2, 3, 4, 5)
Anomaly conditions		Four (a, b, c, d) x 3 damage levels	
Mic. type and channlel config.	Dynamic	Ch. 1-3	Ch. 1-4
	Condenser	Ch. 4, 5	Ch. 5-8
Noise type		Four recortdings	
Normal samples		1,094 samples x 5 models x 5 speed levels	
		91 hours x 5 ch	91 hours x 2 ch-sets
Anomaly samples		324 samples x 5 models x 5 speed levels	
		27 hours x 5 ch	27 hours x 2 ch-sets
Noise samples		Four types (24 hours) per a channel	

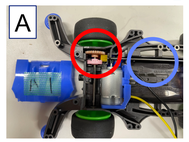
Toy car: Microphone arrangement





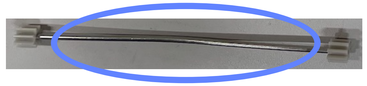
Toy car: Anomaly conditions

	Toy car	Toy train
a	Bent shaft	Obstructing stone
b	Deformed gears	Disjointed railway
c	Melted gears	Broken shaft
d	Damaged wheels	Flat tire



Deformed gears (High)

Bent shaft (High)



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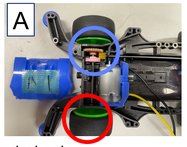
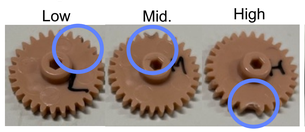
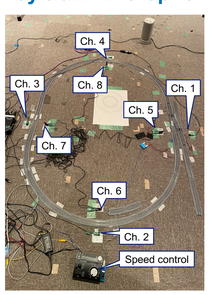


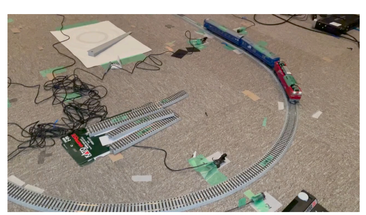
Melted gears



Damaged wheels

Toy train: Microphone arrangement





Toy train: Anomaly conditions

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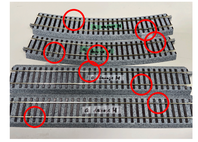
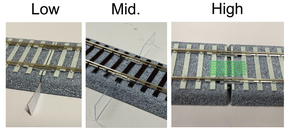
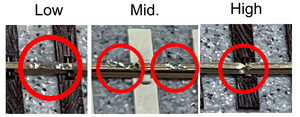


Disjointed railway



Obstructing stone



Toy train: Anomaly conditions

	Toy car	Toy train
a	Bent shaft	Obstructing stone
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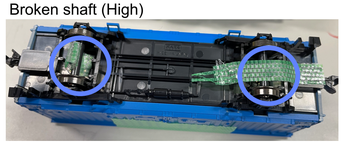
Flat tire (High)

Broken shaft (High)



Recipe (easy template for generating test conditions)

https://github.com/nttcslab/ToyADMOS2-dataset

Example: Making Example Dataset

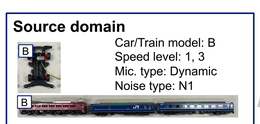
Run the following will create the equivalent benchmark dataset evaluated in the Table 3 of the paper, which is a compatible file-folder structure with the DCASE2021 challenge task 2. This will create dataset folder `your_new_dataset`. **This will take about an hour.**

```
# This creates `clean` dataset.
python mixer.py /path/to/ToyADMOS2 your_new_dataset recipe_benchmark.xlsx clean
# This creates SNR=6dB dataset.
python mixer.py /path/to/ToyADMOS2 your_new_dataset recipe_benchmark.xlsx 6
```

- `recipe_example_car_shift.xlsx` is also another example.
- `recipe_template` is a template, as well as one more example.

Example domain-shift condition

Source domain



Car/Train model: B
Speed level: 1, 3
Mic. type: Dynamic
Noise type: N1

Target domain 1: Model and parts shift

Car/Train model: D
Speed level: 1, 3
Mic. type: Dynamic
Noise type: N1

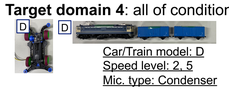
Target domain 2: Operating speed shift

Car/Train model: B
Speed level: 2, 5
Mic. type: Dynamic
Noise type: N1

Target domain 3: Mic. and noise shift

Car/Train model: B
Speed level: 1, 3
Mic. type: Condenser
Noise type: N2

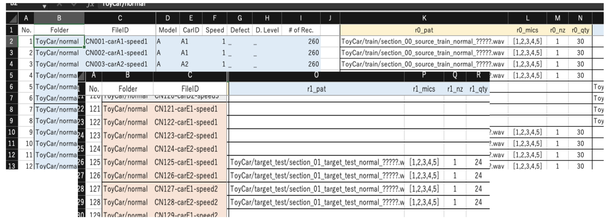
Target domain 4: all of conditions 1 to 3



Car/Train model: D
Speed level: 2, 5
Mic. type: Condenser
Noise type: N2

Recipe (easy template for generating sub-dataset)

You can simply edit the recipe template and run mixer.py to generate a test dataset



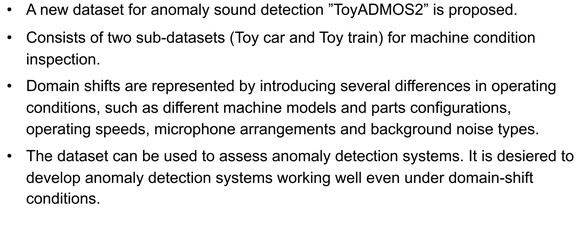
No.	Folder	FileID	Model	CarID	Speed	Defect	D. Level	# of Rec.	r0_pat	r0_mics	r0_nz	r0_qty
1	ToyCar/normal	CN001-car1-speed1	A	A1	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
2	ToyCar/normal	CN002-car1-speed1	A	A1	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
3	ToyCar/normal	CN003-car2-speed1	A	A2	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
4	ToyCar/normal	CN004-car2-speed1	A	A2	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
5	ToyCar/normal	CN005-car3-speed1	A	A3	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
6	ToyCar/normal	CN006-car3-speed1	A	A3	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
7	ToyCar/normal	CN007-car4-speed1	A	A4	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
8	ToyCar/normal	CN008-car4-speed1	A	A4	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
9	ToyCar/normal	CN009-car5-speed1	A	A5	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
10	ToyCar/normal	CN010-car5-speed1	A	A5	1	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
11	ToyCar/normal	CN011-car1-speed2	A	A1	2	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30
12	ToyCar/normal	CN012-car1-speed2	A	A1	2	_	_	260	ToyCar/train/section_00_source_train_normal_?????.wav	[1,2,3,4,5]	1	30

No.	Folder	FileID	r1_pat	r1_mics	r1_nz	r1_qty
121	ToyCar/normal	CN121-car1-speed1				
122	ToyCar/normal	CN122-car1-speed1				
123	ToyCar/normal	CN123-car2-speed1				
124	ToyCar/normal	CN124-car2-speed1				
125	ToyCar/normal	CN125-car1-speed1	ToyCar/target_test/section_01_target_test_normal_?????.w	[1,2,3,4,5]	1	24
126	ToyCar/normal	CN126-car2-speed1	ToyCar/target_test/section_01_target_test_normal_?????.w	[1,2,3,4,5]	1	24
127	ToyCar/normal	CN127-car1-speed2	ToyCar/target_test/section_01_target_test_normal_?????.w	[1,2,3,4,5]	1	24
128	ToyCar/normal	CN128-car2-speed2	ToyCar/target_test/section_01_target_test_normal_?????.w	[1,2,3,4,5]	1	24

```
$ python mixer.py ToyADMOS2 car_shift recipe_example_car_shift.xlsx -6
```

Dataset analisys

Anomaly score for the dataset under clean condition (no background noise) calculated by the DCASE2020 baseline AE



DCASE2021 Challenge T2 results on ToyADMOS2

Rank	System	Source domain results				Target domain results			
		ToyCar		ToyTrain		ToyCar		ToyTrain	
		AUC	pAUC	AUC	pAUC	AUC	pAUC	AUC	pAUC
1	Lopez_IL_task2_4	81.44%	59.05%	77.56%	62.21%	69.97%	60.39%	62.38%	57.78%
2	Morita_SECOM_task2_3	56.47%	53.08%	52.23%	49.60%	66.04%	64.47%	40.47%	49.67%
3	Wilkinghoff_FKIE_task2_3	67.07%	63.05%	70.87%	56.19%	72.83%	63.77%	48.38%	52.39%
4	Kuroyanagi_NU-HDL_task2_3	62.02%	59.69%	71.49%	58.98%	61.38%	52.00%	54.42%	55.27%
5	Sakamoto_Fixstars_task2_1	81.76%	71.53%	66.24%	53.74%	66.46%	63.06%	57.75%	55.29%
21	DCASE2021 baseline task2 AE	76.33%	51.26%	69.89%	55.49%	58.02%	53.42%	67.18%	59.78%

Domain shift

Domain shift

Conclusion

- A new dataset for anomaly sound detection ”ToyADMOS2” is proposed.
- Consists of two sub-datasets (Toy car and Toy train) for machine condition inspection.
- Domain shifts are represented by introducing several differences in operating conditions, such as different machine models and parts configurations, operating speeds, microphone arrangements and background noise types.
- The dataset can be used to assess anomaly detection systems. It is desiered to develop anomaly detection systems working well even under domain-shift conditions.